# TestBoard ADC124 manual ghosting comparison

**Report prepared 20 October 2026.** Six test conditions and 48 array/MUX observations, supplied in Ghosting Tests TestBoard_ADC124.docx. The source does not specify acquisition dates or repeated-press counts.

## Main conclusions

Visiting VMID between external MUX addresses is the strongest observed ghost-suppression measure. With VMID off, Array A shows approximately 90% ghosting on every MUX; Array B shows approximately 30-50%. VMID-on conditions reduce this to no visible ghost or small reported residuals.

ADC conversions while connected to VMID are not essential for substantial suppression in these tests. The wait-only conditions G4 and G5 both suppress the large ghosts. A programmed VMID delay of 0 us still includes a finite reference visit and firmware switching time.

G4 is the fastest reported condition at 9,320 samples/s per channel. Its Array B main signals are 2-5 times smaller than in G3. Manual excitation was uncontrolled, so this does not establish attenuation, but it prevents accepting G4 as a proven improvement in signal fidelity.

G5 is the strongest candidate for a conservative starting configuration: ADC scan, 1 us sensor settle, 5 us VMID settle, no ADC scan at VMID, and 5,620 samples/s per channel. It retains the larger observed Array B signals while reporting only trace residuals. A controlled-input check is needed before choosing production settings.

## Engine verification

The current GUI board profile and firmware both default to **blocking SPI**, not LPSPI. The user reports using the default engine. The report therefore identifies blocking as the verified default; historical runtime engine and firmware identities are not proven by a saved device-status log.

The percentages are visual estimates from manual presses. "No ghost" means none was observed under those conditions; it is not a measured zero or a calibrated rejection limit.

# Test method and physical channel mapping

Each full octo sensor TOP element was pressed manually while observing the adjacent Octo 7 element on the same external MUX. This tests the transition from external MUX address 4 to address 5. MUX 4 also has an Octo 7 sensor at address 6, allowing observation of a second following address. SPI clock was 16 MHz throughout.

| PCB array | MUX / ADC input | Bias resistor | Pressed sensor at address 4 | Observed Octo 7 sensor |
|---|---|---|---|---|
| A | 1 | 470 kohm | PZT1_T | PZT7_B (address 5) |
| A | 2 | 1 Mohm | PZT3_T | PZT7_L (address 5) |
| A | 3 | 470 kohm | PZT5_T | PZT7_C (address 5) |
| A | 4 | 1 Mohm | PZT6_T | PZT7_R (address 5), T(address 6) |
| B | 1 | 1 Mohm | PZT1_T | PZT7_B (address 5) |
| B | 2 | 470 kohm | PZT3_T | PZT7_L (address 5) |
| B | 3 | 1 Mohm | PZT5_T | PZT7_C (address 5) |
| B | 4 | 470 kohm | PZT6_T | PZT7_R (address 5), T(address 6) |

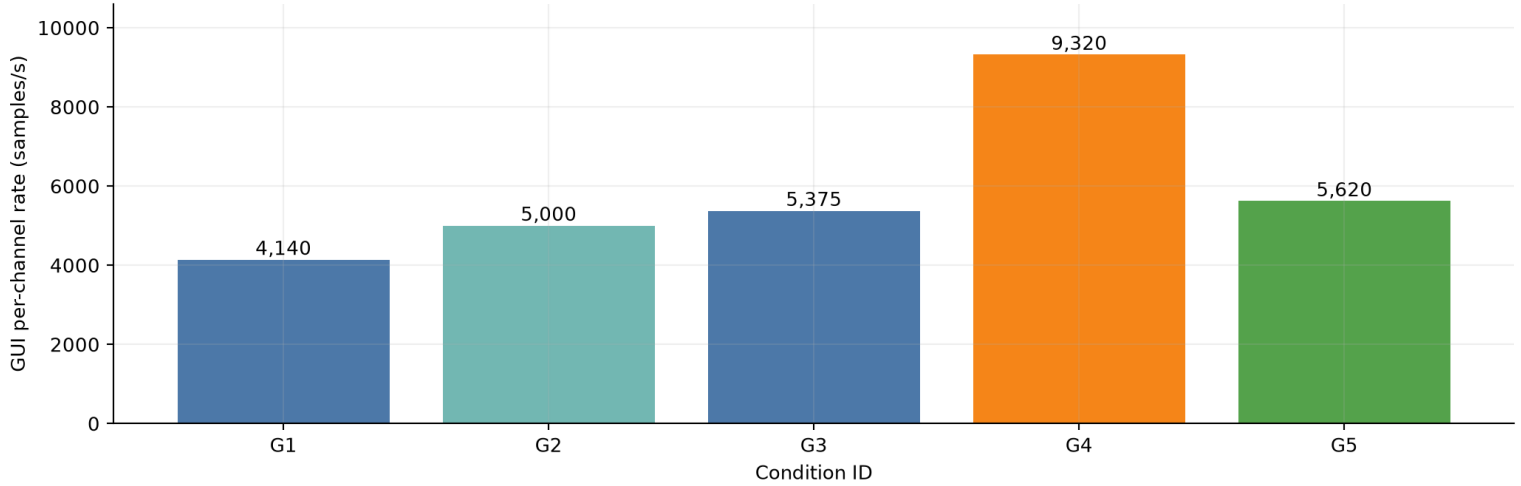
**Terminology:** source "ch4/ch5/ch6" labels are external MUX addresses 4/5/6. MUX number 1-4 identifies a chip and its corresponding ADC input IN1-IN4. These are separate selections. Array 1 = Array A; Array 2 = Array B.

ADC scan completes one array before scanning the other. MUX scan switches both arrays to the same address together and shares the settling wait, then reads the ADCs sequentially. These are the intended firmware semantics; the historical test binary was not archived with the notes.

The source explicitly notes that manual pressure and visual estimates are approximate, and that coupling among elements of the same sensor is difficult to distinguish from mechanical coupling. Strip assignment during this manual session was not specified.

# Test conditions and timing

| ID | Scan order | Sensor settle us | VMID settle us | ADC scan at VMID | GUI per-channel SPS | GUI Sample Interval us | Per-channel period us |
|---|---|---|---|---|---|---|---|
| G0 | ADC | 5 | off | n/a | not stated | not stated | not available |
| G1 | ADC | 5 | 0 | yes | 4140 | 5 | 241.55 |
| G2 | MUX | 5 | 0 | yes | 5000 | 4 | 200.00 |
| G3 | ADC | 1 | 0 | yes | 5375 | 4 | 186.05 |
| G4 | ADC | 1 | 0 | no | 9320 | 2 | 107.30 |
| G5 | ADC | 1 | 5 | no | 5620 | 4 | 177.94 |




**Legend:** bars are the GUI per-channel rates recorded in the notes. G0 has no recorded rate. The channel period is calculated as 1,000,000 / SPS; it is not the GUI Sample Interval.

The user confirmed the values came from the GUI Sampling Rate panel. Current GUI code displays Sample Interval from the MCU average time per serialized sample, while per-channel rate comes from measured sweep timing. With 50 acquired channels, the reported 2-5 us intervals are consistent with approximate average code spacing; one sensor is revisited every 107-242 us in G1-G5.

# Ghosting observations across all conditions

| Condition | A MUX 1 | A MUX 2 | A MUX 3 | A MUX 4 | B MUX 1 | B MUX 2 | B MUX 3 | B MUX 4 |
|---|---|---|---|---|---|---|---|---|
| G0 | ~90% | ~90% | ~90% | ~90% | ~50% | ~30% | ~30% | ~30% |
| G1 | ND | ND | ND | ND | ND | ND | <5% | ND |
| G2 | ND | ND | <1% N | ND | ND | ND | <5% | ND |
| G3 | ND | ND | ND | ND | ND | ND | <1% | <1% N |
| G4 | ND | ND | ND | ND | <2% | <1% | <1% | <1% N |
| G5 | ND | ND | ND | ND | ND | ND | <1% | <1% N |

**Legend:** ND = source says "No ghost"; no numerical detection threshold was measured. ~ = approximate visual fraction. <1%, <2% and <5% reproduce the reported visual estimates, not statistical upper confidence bounds. N = source also reports a changed noise pattern.

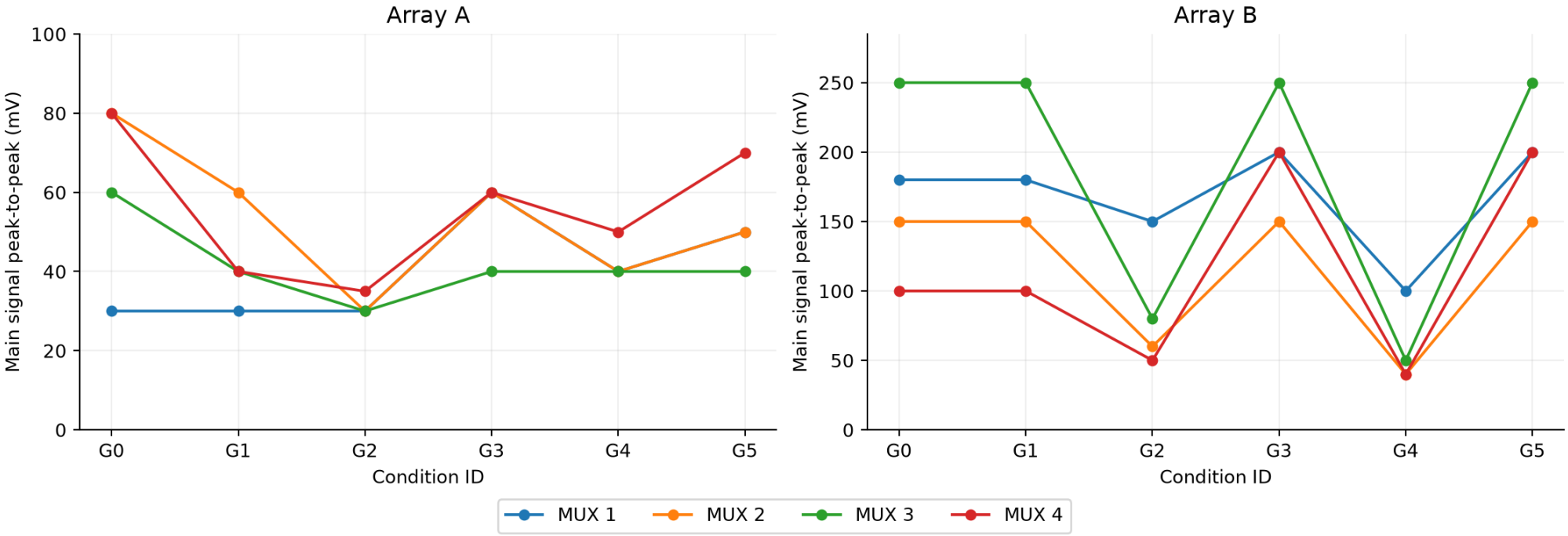
## What the comparison establishes

G0 to G1 is the closest VMID-off/on comparison: both use ADC scan and 5 us sensor settling. Array A changes from approximately 90% on all four MUXes to ND. Array B changes from approximately 30-50% to ND on three MUXes and <5% on MUX 3. VMID adds switching and conversion time, so this comparison establishes the benefit of the complete VMID visit rather than isolating the reference capacitor.

Array B / MUX 3 is the persistent residual path: a small ghost is noted in every VMID-on condition. Array A / MUX 3 has a trace observation only in G2. Source remarks about a changed noise pattern on B / MUX 4 should remain separate from a measured ghost amplitude.

With VMID off, A / MUX 4 also shows ghosting on the second following sensor at external address 6. B / MUX 4 shows hardly any ghost there. This is consistent with a longer scan-memory tail on A, although the second-address magnitude was not quantified.

# Main signal amplitudes and the fastest setting




**Legend:** colour = MUX number. Each point is the source main-signal peak-to-peak amplitude in mV, not ghost amplitude or noise. Condition IDs are categorical; lines only help follow a MUX across tests. Presses were not identical or force-controlled.

| Array B path | G3 with ADC scan mV p-p | G4 no ADC scan mV p-p | G4 / G3 | G5 with 5 us VMID mV p-p |
|---|---|---|---|---|
| MUX 1 | 200.0 | 100.0 | 0.50 | 200.0 |
| MUX 2 | 150.0 | 40.0 | 0.27 | 150.0 |
| MUX 3 | 250.0 | 50.0 | 0.20 | 250.0 |
| MUX 4 | 200.0 | 40.0 | 0.20 | 200.0 |

The smaller G4 signals may result from different presses, acquisition-history effects or reduced measured response. The notes cannot distinguish these explanations. A lower visible ghost fraction at a smaller signal can also approach the ADC/noise floor; speed alone is insufficient to choose G4.

# Effects of scan order settling and VMID conversions

| Comparison | Changed setting | Per-channel rate before / after | Rate change | Period change us |
|---|---|---|---|---|
| G1 to G2 | ADC scan to MUX scan | 4,140 / 5,000 | +20.8% | -41.55 |
| G1 to G3 | Sensor settle 5 to 1 us | 4,140 / 5,375 | +29.8% | -55.50 |
| G3 to G4 | Remove ADC scan at VMID | 5,375 / 9,320 | +73.4% | -78.75 |
| G4 to G5 | Add 5 us wait at VMID | 9,320 / 5,620 | -39.7% | +70.64 |
| G3 to G5 | Replace VMID ADC scan by 5 us wait | 5,375 / 5,620 | +4.6% | -8.11 |

**Scan order G1 to G2.** Shared MUX settling increases the recorded rate by 20.8% at 5 us sensor settle. Both retain strong suppression, but G2 adds a trace observation on A / MUX 3 and still reports <5% on B / MUX 3. The notes do not test MUX scan at 1 us or without VMID conversions.

**Sensor settle G1 to G3.** Reducing the programmed sensor wait to 1 us increases rate by 29.8% without a large observed ghost returning. B / MUX 3 changes from <5% to <1%, while B / MUX 4 has a trace noise-pattern observation. These qualitative observations do not establish that shorter settling improves analog accuracy.

**VMID conversions G3 to G4.** Removing them increases rate by 73.4%. Array A remains ND; Array B has reported residuals below 1-2%. The concurrent decrease in major-signal amplitudes prevents interpreting this as a demonstrated fidelity improvement.

**VMID wait G4 to G5.** Rate decreases by 39.7%, but observed Array B signals return to 150-250 mV on MUXes 1-3 and 200 mV on MUX 4. B / MUXes 1-2 become ND. G5 is also 4.6% faster than G3 and has the same reported ghost categories, making wait-only VMID a credible alternative to VMID conversions.

Calculated G4/G5 periods are 107.30 and 177.94 us, a difference of 70.64 us. If all 50 sensor routes are scanned over seven addresses per array, ADC order has 14 VMID visits: 14 x 5 us = 70 us. This close match supports the intended timing interpretation; it is not an independent GPIO timing measurement.

# Electrical interpretation of the array differences

## The results support scan memory as a major contributor

A previous sensor can leave charge on a MUX common node and associated buffer/input capacitances. Connecting the next high-impedance sensor can redistribute that charge and shift the next reading. A VMID visit drives the shared path toward a known reference before the next sensor is connected. This is a plausible explanation for the strong address-4 to address-5 dependence and the improvement with wait-only VMID.

The data do not locate the entire effect in one component. MUX parasitics, OPA365 response, sensor/cable capacitance and ADC acquisition history can contribute. Removing VMID ADC conversions shows they are not necessary for substantial suppression under the tested conditions; it does not prove the ADC contributes nothing.

| Typical datasheet parameter at 3.3 V | Array A TMUX1108 | Array B TMUX1308A |
|---|---|---|
| On capacitance | 65 pF | 11 pF |
| On resistance | 4 ohm | 120 ohm |
| Sensor-side off-state leakage magnitude at 25 C | 1 pA | 1 nA |

The larger stated on capacitance of TMUX1108 makes greater capacitive memory on A plausible. Its lower off-state leakage also preserves stored charge more effectively. TMUX1308A can have less capacitive memory while losing more stored sensor voltage during long isolation. Thus strong ghosting on A and stronger long-hold discharge on B can coexist. This is an interpretation consistent with both test sets, not a component-level proof. [2, 3]

These are typical manufacturer values under their specified test conditions, not measured PCB capacitance or leakage. The leakage test voltages differ. On resistance alone does not predict ghosting into a capacitive, high-impedance sensor path. The earlier strip-exchange results do not establish which strip assignment was used in this manual session.

## The resistor comparison is inconclusive

Both 470 kohm and 1 Mohm paths on Array A show approximately 90% ghosting with VMID off. On B, only MUX 1 is estimated at 50%; the other three are 30%. These observations do not isolate a consistent resistor-value effect because MUX position, individual sensor and applied pressure vary together.

# Configuration choices and required confirmation

## Use G5 as the conservative validation starting point

Keep SPI at 16 MHz, ADC scan, 1 us sensor settle, VMID enabled with a 5 us wait, and no ADC conversions at VMID. The recorded rate is 5,620 samples/s per channel. Its manual results combine large main signals and only trace residuals on B / MUXes 3-4. This is a provisional test configuration, not a validated production limit.

## Treat G4 as the speed candidate

G4 reaches 9,320 samples/s per channel with 1 us sensor settle and no explicit VMID delay or VMID conversions. Before selecting it, verify that the desired signal amplitude, waveform and integrated response are preserved under identical excitation. A 0 us programmed wait leaves finite analog dwell from switching and firmware execution.

## Run the following comparisons with controlled excitation

**1.** Compare G3, G4 and G5 using a repeatable actuator trajectory or calibrated charge injection that preserves representative sensor impedance. Record the pressed path and all other paths together. Use at least ten events per condition and reverse or randomize condition order to expose drift.

**2.** Measure positive and negative ghost waveforms relative to each victim baseline, including delayed responses at following addresses. Compute a consistent ghost-to-source amplitude ratio, main-signal retention and trial-to-trial variation. Record pre-event noise to establish a detection limit.

**3.** Repeat selected tests with scan order reversed or with the stimulus address changed. A response that follows read order supports electrical scan memory; one that follows physical neighbours may include mechanical coupling. Fixed capacitors or unconnected dummy channels help separate the two.

**4.** Sweep wait-only VMID delay through 0, 0.5, 1, 2 and 5 us where firmware resolution permits. Test the MUX scan equivalent of G5. Log actual requested and achieved timing, active SPI engine, firmware identity, all routes and strip assignment. Keep excitation and conditioning history fixed.

# Measurement limits units and references

Manual presses vary in force, displacement and speed. The document contains observations and main-signal peak-to-peak values, without raw traces, event counts, uncertainty intervals or a quantitative ghost-detection floor. It cannot establish statistically significant differences among sub-percent categories, a complete 50-by-50 ghost matrix, noise SD/PSD, or calibrated signal gain.

At the nominal 3.3 V reference, one 12-bit ADC count is 0.805664 mV. A visually estimated 1% ghost on a 40 mV main signal corresponds to only 0.4 mV, about 0.50 count; on a 250 mV signal it corresponds to 2.5 mV, about 3.10 counts. These products illustrate scale only. The source does not establish that its percentages use identical peak-to-peak windows, so they are not validated voltage bounds.

The result tables use ND for an absent visible ghost, never zero percent. Statements such as <1% retain their original observational status. Main-signal peak-to-peak amplitudes must not be relabelled as ghost amplitudes or noise levels. G0 sampling rate is missing and remains unfilled.

## Sources and verification

[1] Ghosting Tests TestBoard_ADC124.docx, supplied manual notes: one mapping table, six result tables, 48 result rows. The appendix reproduces every observation and main-signal value. Source SHA-256: f276dee2195b207055b791ef43da23ab801acb40bd500c70a9526ec810dfd9d9.

[2] Texas Instruments, TMUX1108 datasheet, sections 6.6 and 7.3.6. TI TMUX1108 datasheet. [3] Texas Instruments, TMUX1308A datasheet, sections 6.6 and 7.7. TI TMUX1308A datasheet.

[4] Repository checked 20 October 2026: config/boards/profiles/testboard_adc124.json sets spi_engine.default to blocking; Arduino_Sketches/TestBoard_ADC124/src/main.cpp initializes SpiEngine::BLOCKING. This verifies defaults only. Current firmware identifies wait-only VMID; the source test sequence includes earlier VMID-conversion behaviour, so one present-day binary cannot describe every historical row.

[5] data_processing/timing_display.py, update_timing_display: Sample Interval displays the MCU average sample time; per-channel rate uses actual sweep timing. User clarification confirms the document values were copied from that GUI panel. No board configuration was changed or hardware test run for this report.

# Appendix G0 complete manual observations

**Parameters:** ADC scan, sensor settle 5 us, VMID off. SPI 16 MHz. Rate not recorded.

| Array | MUX / bias | Pressed external address 4 | Observed external address 5 or 6 | Source ghost observation | Main signal mV p-p |
|---|---|---|---|---|---|
| A | 1 / 470k | PZT1_T | PZT7_B (addr 5) | Very strong ghost (90%) | 30 |
| A | 2 / 1M | PZT3_T | PZT7_L (addr 5) | Very strong ghost (90%) | 80 |
| A | 3 / 470k | PZT5_T | PZT7_C (addr 5) | Very strong ghost (90%) | 60 |
| A | 4 / 1M | PZT6_T | PZT7_R (addr 5), T(addr 6) | Very strong ghost (90%). Ghost on second channel. | 80 |
| B | 1 / 1M | PZT1_T | PZT7_B (addr 5) | Strong ghost (50%) | 180 |
| B | 2 / 470k | PZT3_T | PZT7_L (addr 5) | Ghost (30%) | 150 |
| B | 3 / 1M | PZT5_T | PZT7_C (addr 5) | Ghost (30%) | 250 |
| B | 4 / 470k | PZT6_T | PZT7_R (addr 5), T(addr 6) | Ghost (30%). Hardly any ghost on second channel. | 100 |

The source observations are reproduced without assigning numerical zeros to "No ghost". MUX number identifies the same-numbered ADC input. "Second channel" in the MUX 4 remarks means the second following external address, address 6, rather than ADC IN2.

Conclusion: strong ghosts occur on every observed path, with a longer visible tail on Array A / MUX 4. The signal amplitudes vary across manual presses, so percentages should be treated as approximate.

# Appendix G1 complete manual observations

**Parameters:** ADC scan, sensor settle 5 us, VMID 0 us, VMID ADC scan on. SPI 16 MHz. GUI rate 4,140 samples/s per channel; GUI Sample Interval 5 us.

| Array | MUX / bias | Pressed external address 4 | Observed external address 5 or 6 | Source ghost observation | Main signal mV p-p |
|---|---|---|---|---|---|
| A | 1 / 470k | PZT1_T | PZT7_B (addr 5) | No ghost | 30 |
| A | 2 / 1M | PZT3_T | PZT7_L (addr 5) | No ghost | 60 |
| A | 3 / 470k | PZT5_T | PZT7_C (addr 5) | No ghost | 40 |
| A | 4 / 1M | PZT6_T | PZT7_R (addr 5), T(addr 6) | No ghost | 40 |
| B | 1 / 1M | PZT1_T | PZT7_B (addr 5) | No ghost | 180 |
| B | 2 / 470k | PZT3_T | PZT7_L (addr 5) | No ghost | 150 |
| B | 3 / 1M | PZT5_T | PZT7_C (addr 5) | Very small ghost (<5%) | 250 |
| B | 4 / 470k | PZT6_T | PZT7_R (addr 5), T(addr 6) | No ghost | 100 |

The source observations are reproduced without assigning numerical zeros to "No ghost". MUX number identifies the same-numbered ADC input. "Second channel" in the MUX 4 remarks means the second following external address, address 6, rather than ADC IN2.

Conclusion: adding the VMID visit strongly suppresses the G0 ghosts at unchanged 5 us sensor settle. B / MUX 3 retains a small reported residual below 5%. Zero programmed VMID wait includes the additional time spent performing ADC conversions at VMID.

# Appendix G2 complete manual observations

**Parameters:** MUX scan, sensor settle 5 us, VMID 0 us, VMID ADC scan on. SPI 16 MHz. GUI rate 5,000 samples/s per channel; GUI Sample Interval 4 us.

| Array | MUX / bias | Pressed external address 4 | Observed external address 5 or 6 | Source ghost observation | Main signal mV p-p |
|---|---|---|---|---|---|
| A | 1 / 470k | PZT1_T | PZT7_B (addr 5) | No ghost | 30 |
| A | 2 / 1M | PZT3_T | PZT7_L (addr 5) | No ghost | 30 |
| A | 3 / 470k | PZT5_T | PZT7_C (addr 5) | Almost no ghost (<1%, noise pattern changes) | 30 |
| A | 4 / 1M | PZT6_T | PZT7_R (addr 5), T(addr 6) | No ghost | 35 |
| B | 1 / 1M | PZT1_T | PZT7_B (addr 5) | No ghost | 150 |
| B | 2 / 470k | PZT3_T | PZT7_L (addr 5) | No ghost | 60 |
| B | 3 / 1M | PZT5_T | PZT7_C (addr 5) | Very small ghost (<5%) | 80 |
| B | 4 / 470k | PZT6_T | PZT7_R (addr 5), T(addr 6) | No ghost | 50 |

The source observations are reproduced without assigning numerical zeros to "No ghost". MUX number identifies the same-numbered ADC input. "Second channel" in the MUX 4 remarks means the second following external address, address 6, rather than ADC IN2.

Conclusion: shared MUX settling raises the recorded rate versus G1. A / MUX 3 has a trace observation and B / MUX 3 remains below 5%. Lower main amplitudes on several paths cannot be assigned to scan mode without repeatable excitation.

# Appendix G3 complete manual observations

**Parameters:** ADC scan, sensor settle 1 us, VMID 0 us, VMID ADC scan on. SPI 16 MHz. GUI rate 5,375 samples/s per channel; GUI Sample Interval 4 us.

| Array | MUX / bias | Pressed external address 4 | Observed external address 5 or 6 | Source ghost observation | Main signal mV p-p |
|---|---|---|---|---|---|
| A | 1 / 470k | PZT1_T | PZT7_B (addr 5) | No ghost | 60 |
| A | 2 / 1M | PZT3_T | PZT7_L (addr 5) | No ghost | 60 |
| A | 3 / 470k | PZT5_T | PZT7_C (addr 5) | No ghost | 40 |
| A | 4 / 1M | PZT6_T | PZT7_R (addr 5), T(addr 6) | No ghost | 60 |
| B | 1 / 1M | PZT1_T | PZT7_B (addr 5) | No ghost | 200 |
| B | 2 / 470k | PZT3_T | PZT7_L (addr 5) | No ghost | 150 |
| B | 3 / 1M | PZT5_T | PZT7_C (addr 5) | Very small ghost (<1%) | 250 |
| B | 4 / 470k | PZT6_T | PZT7_R (addr 5), T(addr 6) | Almost no ghost (<1%, noise pattern changes) | 200 |

The source observations are reproduced without assigning numerical zeros to "No ghost". MUX number identifies the same-numbered ADC input. "Second channel" in the MUX 4 remarks means the second following external address, address 6, rather than ADC IN2.

Conclusion: 1 us sensor settle retains strong suppression and raises rate versus G1. The two residual observations are on B / MUXes 3-4. This establishes useful behaviour in this manual test, not full analog settling or calibrated accuracy.

# Appendix G4 complete manual observations

**Parameters:** ADC scan, sensor settle 1 us, VMID 0 us, VMID ADC scan off. SPI 16 MHz. GUI rate 9,320 samples/s per channel; GUI Sample Interval 2 us.

| Array | MUX / bias | Pressed external address 4 | Observed external address 5 or 6 | Source ghost observation | Main signal mV p-p |
|---|---|---|---|---|---|
| A | 1 / 470k | PZT1_T | PZT7_B (addr 5) | No ghost | 40 |
| A | 2 / 1M | PZT3_T | PZT7_L (addr 5) | No ghost | 40 |
| A | 3 / 470k | PZT5_T | PZT7_C (addr 5) | No ghost | 40 |
| A | 4 / 1M | PZT6_T | PZT7_R (addr 5), T(addr 6) | No ghost | 50 |
| B | 1 / 1M | PZT1_T | PZT7_B (addr 5) | Very small ghost (<2%) | 100 |
| B | 2 / 470k | PZT3_T | PZT7_L (addr 5) | Very small ghost (<1%) | 40 |
| B | 3 / 1M | PZT5_T | PZT7_C (addr 5) | Very small ghost (<1%) | 50 |
| B | 4 / 470k | PZT6_T | PZT7_R (addr 5), T(addr 6) | Almost no ghost (<1%, noise pattern changes) | 40 |

The source observations are reproduced without assigning numerical zeros to "No ghost". MUX number identifies the same-numbered ADC input. "Second channel" in the MUX 4 remarks means the second following external address, address 6, rather than ADC IN2.

Conclusion: substantial suppression survives without VMID ADC conversions, at the highest recorded rate. All four B paths have trace observations and smaller main signals than G3. Confirm main-signal preservation with a controlled input before adopting this condition.

# Appendix G5 complete manual observations

**Parameters:** ADC scan, sensor settle 1 us, VMID 5 us, VMID ADC scan off. SPI 16 MHz. GUI rate 5,620 samples/s per channel; GUI Sample Interval 4 us.

| Array | MUX / bias | Pressed external address 4 | Observed external address 5 or 6 | Source ghost observation | Main signal mV p-p |
|---|---|---|---|---|---|
| A | 1 / 470k | PZT1_T | PZT7_B (addr 5) | No ghost | 50 |
| A | 2 / 1M | PZT3_T | PZT7_L (addr 5) | No ghost | 50 |
| A | 3 / 470k | PZT5_T | PZT7_C (addr 5) | No ghost | 40 |
| A | 4 / 1M | PZT6_T | PZT7_R (addr 5), T(addr 6) | No ghost | 70 |
| B | 1 / 1M | PZT1_T | PZT7_B (addr 5) | No ghost | 200 |
| B | 2 / 470k | PZT3_T | PZT7_L (addr 5) | No ghost | 150 |
| B | 3 / 1M | PZT5_T | PZT7_C (addr 5) | Very small ghost (<1%) | 250 |
| B | 4 / 470k | PZT6_T | PZT7_R (addr 5), T(addr 6) | Almost no ghost (<1%, noise pattern changes) | 200 |

The source observations are reproduced without assigning numerical zeros to "No ghost". MUX number identifies the same-numbered ADC input. "Second channel" in the MUX 4 remarks means the second following external address, address 6, rather than ADC IN2.

Conclusion: a 5 us reference wait without VMID ADC conversions produces larger observed main signals on B and removes its MUX 1-2 visible residuals relative to G4. It is a conservative validation candidate; tiny remaining observations are not calibrated rejection limits.